Central elements that cannot be distinguished generators

Human Compression project: finite enumeration supplement

Working supplement; distinct review pending

The full retained proofs of enumeration by prefix stabilizers and by double cosets remain in force. This note supplies one additional exact pruning rule. It uses the elementary derived subgroup, not a heuristic about the frequency of possible completions.

Lemma 1. *Let $G = \langle t_0, \ldots, t_{r-1} \rangle$ be a string C-group. No distinguished generator belongs to $Z(G) \cap G'$.*

Proof. Suppose t_i is central, and set $H = \langle t_j : j \neq i \rangle$. The intersection property gives $\langle t_i \rangle \cap H = 1$. Since t_i has order two and is central, multiplication defines an isomorphism $\langle t_i \rangle \times H \longrightarrow G$. Every commutator in this direct product has trivial first coordinate. Thus $G' \leq H$, while $t_i \notin H$. It follows that $t_i \notin G'$, proving the assertion. □

Consequently every candidate lying in the characteristic subgroup $Z(G) \cap G'$ may be discarded before forming stabilizer orbits. The remaining candidate domain is invariant under every automorphism of G, and hence under every prefix stabilizer. Completeness and uniqueness in the retained orbit proof are unaffected: no valid marked generating tuple contains an excluded entry.

The same rule applies to the retained double-coset implementation. For a candidate centralizer Q, remove its involution classes whose representatives lie in $Z(G) \cap G'$. Membership is constant on each such class, because Q acts on G by automorphisms. The retained normal closure of the remaining representatives in Q then generates all remaining eligible elements. If this subgroup together with the old prefix fails to generate G, the branch still cannot have a completion. Every valid completion must use only the remaining candidates. Neither argument assumes that the diagram is connected.

The implementation computes $Z(G) \cap G'$ once, by exact GAP group operations, and changes only the class filter in the existing pool construction. It retains the same automorphism-action guard, subgroup intersection tests, full rank bound, tuple records and duality tests. The preceding lemma is elementary and no novelty is claimed. The group-library calculations remain software premises; there is no Lean claim and no new whole-original recovery credit.